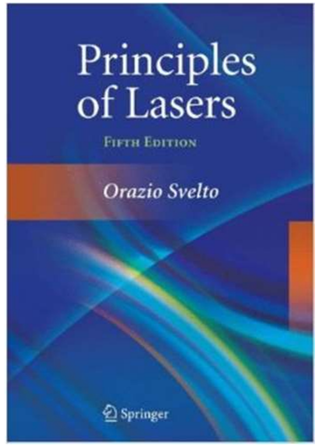
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch2-1
Open Cavity & Gaussian Beam

Dr. Dan Luo(罗丹)
Email: luod@sustech.edu.cn
Office: Room 331 at College of Engineering South Tower
Tel: 0755-88018552

Chapter 2: Open Cavity & Gaussian Beam

❖ 2.1 General Questions for Optical Cavity
- Mode & cavity
- Cavity loss
- Classification of optical cavity

❖ 2.2 Matrix Formulation of Geometrical Optics
- ABCD law
- Stable condition for coaxial cavity

2.1 General Question for Optical Cavity

❖ Mode & Cavity

- Optical mode is the photon state that can be distinguished inside the optical cavity
- The resonance mode is fixed once the cavity is given
- Mode theory for optical cavity is the laser mode theory
- Characteristics of mode: energy distribution, frequency, loss, angle of divergence…

The resonance condition: After one period, it can return to the starting point with the same phase (= n*2π)

$$\Delta\Phi = \frac{2\pi}{\lambda_q} 2L' = q2\pi \Rightarrow L' = q\frac{\lambda_q}{2}$$

$$\nu = c/\lambda$$

$$\nu = q\frac{c}{2L'}$$

L' : optical length

λ_q: wavelength in vacuum

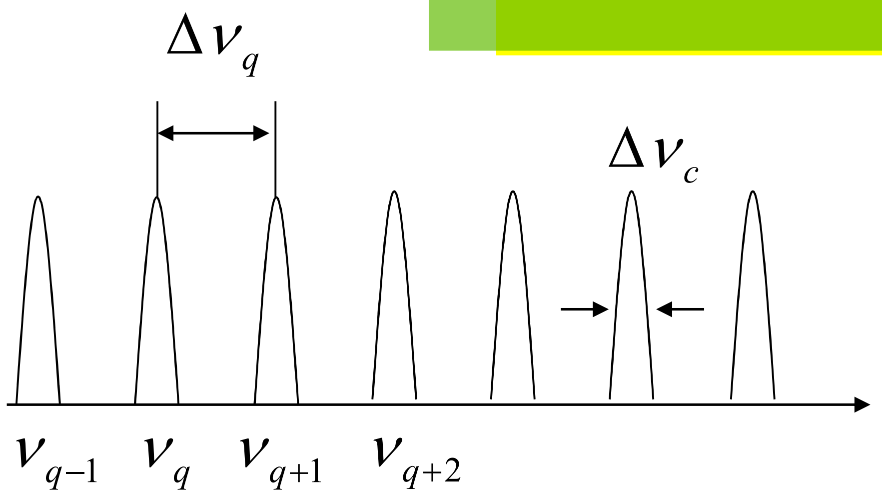
When the cavity is filled with material with refractive index *n*, then

$$\left.\begin{aligned} L' &= nL \\ \nu_q &= q\frac{c}{2L'} = q\frac{c}{2nL} \end{aligned}\right\} \quad L = q\frac{\lambda_q'}{2}$$

L is the cavity length

λ' is the resonance wavelength in materials

$$\Delta\nu_q = \nu_{q+1} - \nu_q = \frac{c}{2L'}$$



(1) For L=10 cm, n=1,

$$\Delta\nu_q = 1.5\times10^9\,\text{Hz}$$

(2) Smaller cavity length, the larger longitudinal mode spacing

Cavity Loss

1. Transmittance Loss (due to low reflection) $\delta = -\frac{1}{2}\ln r_1 r_2$
2. Diffraction Loss (due to diffraction)
3. Geometrical loss (due to angle)
4. Other Loss (material, *Q*-switch element…)

$$\left.\begin{aligned} &\delta = \delta_1 + \delta_2 + \delta_3 + \delta_4 + \ldots \\ &I_1 = I_0 e^{-2\delta} \end{aligned}\right\}$$

$$\boxed{\delta = \frac{1}{2}\ln\frac{I_0}{I_1} \quad \text{The average loss per cycle}}$$

$$\boxed{\delta' = \frac{1}{2}\frac{I_0 - I_1}{I_0} \quad \text{The percentage loss per cycle}}$$

When the cavity loss is very small

$$2\delta' = \frac{I_0 - I_1}{I_0} = \frac{I_0 - I_0 e^{-2\delta}}{I_0} = 1 - \boxed{e^{-2\delta}} \approx 1 - (1 - 2\delta) = 2\delta$$

Parameters related to cavity loss

(1) The average photon lifetime: time needed for the intensity decrease to 1/e

$$I_m = I_0 e^{-2\delta m} \xrightarrow{m = t/(2L'/c)} I_m = I_0 e^{-\frac{2\delta c t}{2L'}} \longrightarrow I(t) = I_0 e^{-\frac{t}{\tau_R}}$$

$$\frac{\delta c}{L'} = \frac{1}{\tau_R} \Rightarrow \boxed{\tau_R = L'/\delta c}$$

optical length

density of photon number

$$I \propto N \qquad N(t) = N_0 e^{-\frac{t}{\tau_R}}$$

$$-dN = \frac{N_0}{\tau_R} e^{-\frac{t}{\tau_R}} dt$$

$$\boxed{\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}}$$

$$\bar{t} = \frac{1}{N_0}\int(-dN)t = \frac{1}{N_0}\int_0^\infty t\left(\frac{N_0}{\tau_R}\right)e^{-\frac{t}{\tau_R}}dt = \left(\frac{1}{\tau_R}\right)\int_0^\infty t e^{-\frac{t}{\tau_R}}dt = \tau_R$$

Parameters related to cavity loss

(2) Q-factor for passive cavity

$$Q = 2\pi\nu \frac{\text{Total energy in the cavity（}\xi\text{）}}{\text{Energy loss per unit time（}P\text{）}}$$

$$\xi = Nh\nu V$$

$$P = -h\nu V \frac{dN}{dt}$$

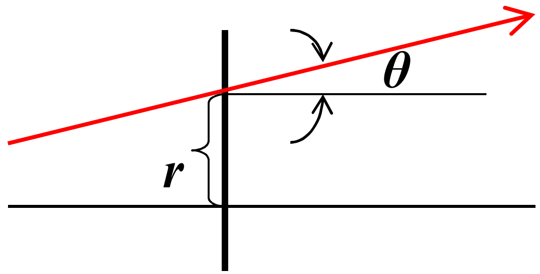
$$N = N_0 e^{-\frac{t}{\tau_R}} \longrightarrow Q = 2\pi\nu \frac{L'}{\delta c} = \omega\tau_R$$

$$\boxed{Q \propto \tau_R \propto \frac{1}{\delta}}$$

The smaller the cavity loss, the higher the Q-factor of the cavity

2.2 Matrix Formulation of Geometrical Optics

1. Formulation of Geometrical Optics（ABCD matrix）

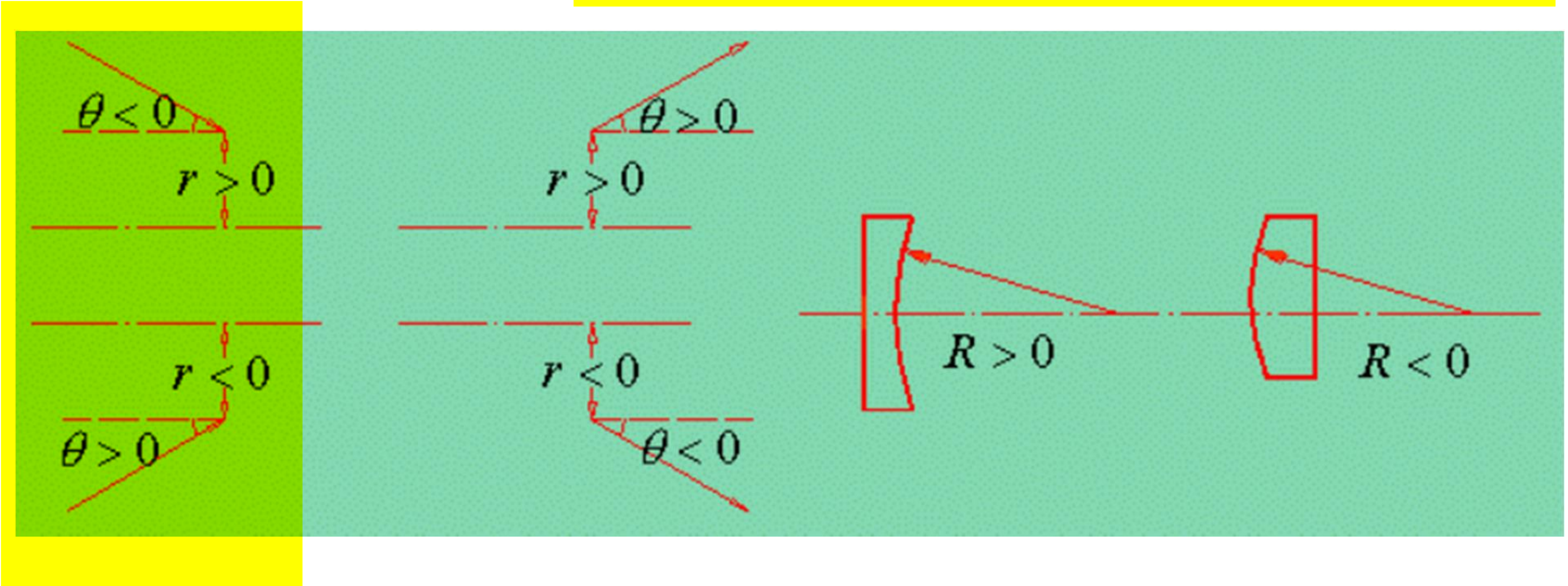


A ray can be described by

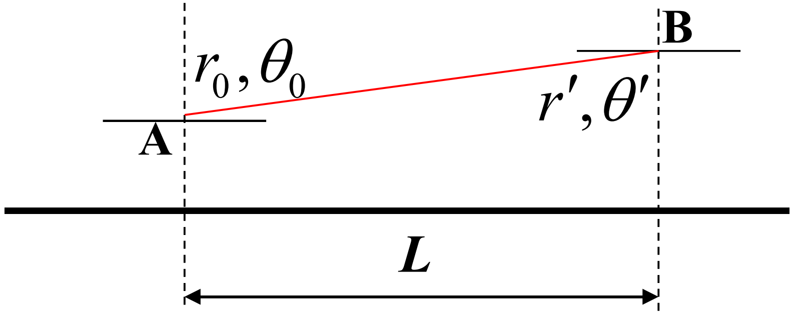
r — distance with optical axis

θ — angle with optical axis

Small angle (paraxial-ray) $tg\theta \approx sin\theta \approx \theta$



2. The matrix in free space



A: $r_{0,}\ \theta_0$

B: $r'_{,}\ \theta'$

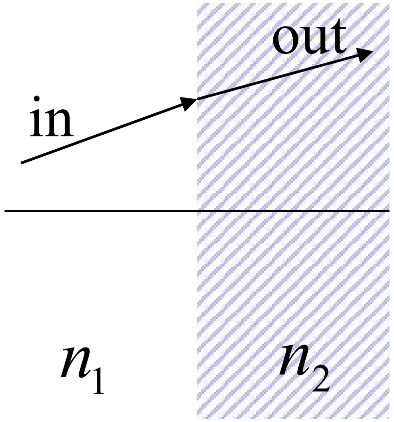
$\Rightarrow$

$$r' = r_0 + L\theta_0$$
$$\theta' = \theta_0$$

Free Space Ray Matrix

$$\begin{pmatrix} r' \\ \theta' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = T_L \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} \Rightarrow \boxed{T_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}}$$

3. Interface between air and dielectric (refractive index n_2)



in $\quad r_0, \theta_0$

out $\quad r', \theta'$

$\Rightarrow$

$$r' = r_0$$
$$\theta' = \frac{n_1}{n_2}\theta_0$$

$\Rightarrow$

$$\boxed{T_L = \begin{pmatrix} 1 & 0 \\ 0 & n_1/n_2 \end{pmatrix}}$$

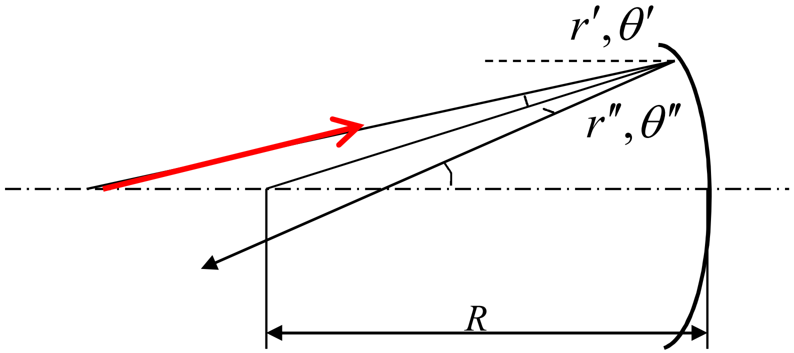
$n_1\theta_0 = n_2\theta'$ (Snell's law)

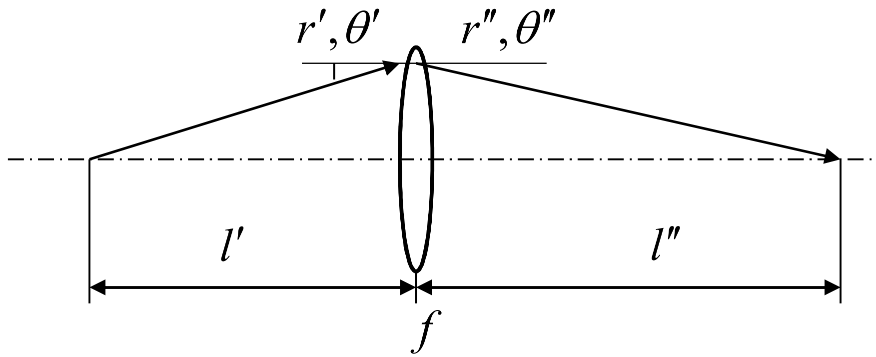
4. The matrix for spherical mirror (and lens)

$$r'' = r' \qquad \theta'' = \theta' - \frac{2}{R} r' \quad \Rightarrow \quad \begin{pmatrix} r'' \\ \theta'' \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r' \\ \theta' \end{pmatrix} \quad \Rightarrow \quad \boxed{T_R = \begin{pmatrix} 1 & 0 \\ -2/R & 1 \end{pmatrix}}$$

Planer mirror

$$T_R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$





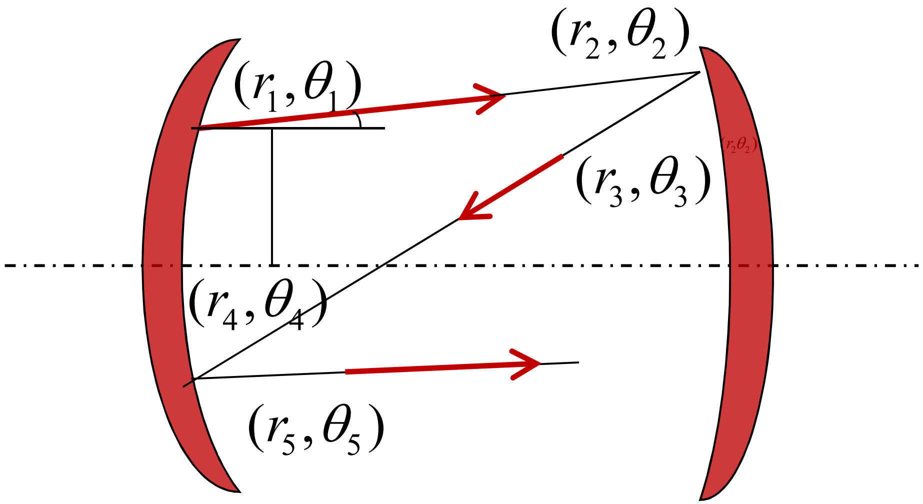
$$r'' = r' \qquad \theta' = r'/l'$$

$$-\theta'' = r''/l''$$

$$\theta' - \theta'' = r'\left(\frac{1}{l'} + \frac{1}{l''}\right) = r'\frac{1}{f}$$

$$\Rightarrow \quad r'' = r' \qquad \theta'' = -\frac{1}{f} r' + \theta' \quad \Rightarrow \quad \boxed{T_f = \begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix} \quad f = \frac{R}{2}}$$

The two are the same



The first travel $$\begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} = T_L \begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

The second travel $$\begin{pmatrix} r_3 \\ \theta_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_2} & 1 \end{pmatrix} \begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} = T_L \begin{pmatrix} r_2 \\ \theta_2 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_2} & 1 \end{pmatrix}$$

The third travel $$\begin{pmatrix} r_4 \\ \theta_4 \end{pmatrix} = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} r_3 \\ \theta_3 \end{pmatrix} = T_L \begin{pmatrix} r_3 \\ \theta_3 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

The fourth travel $$\begin{pmatrix} r_5 \\ \theta_5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_1} & 1 \end{pmatrix} \begin{pmatrix} r_4 \\ \theta_4 \end{pmatrix} = T_L \begin{pmatrix} r_4 \\ \theta_4 \end{pmatrix} \Rightarrow T_L = \begin{pmatrix} 1 & 0 \\ -\frac{2}{R_1} & 1 \end{pmatrix}$$

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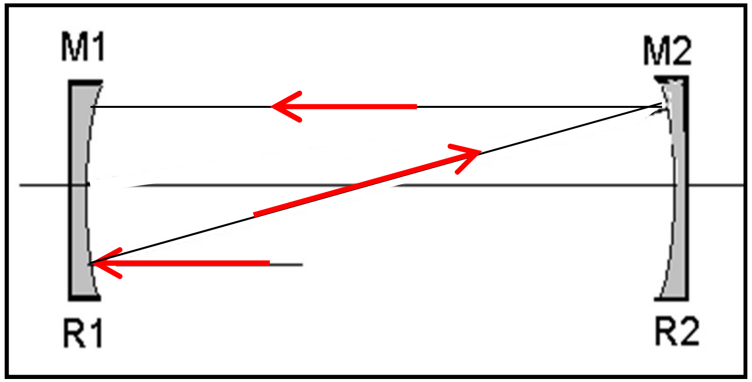
5. ABCD Matrix Application (resonance inside spherical mirror)

球面镜腔中往返一周的光线矩阵（简称往返矩阵）



$$\begin{pmatrix} r_1 \\ \theta_1 \end{pmatrix} = T_{R_1} T_L T_{R_2} T_L \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

$$= T \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2/R_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2/R_2 & 1 \end{pmatrix} \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}$$

$$A = 1 - \frac{2L}{R_2} \qquad B = 2L\left(1 - \frac{L}{R_2}\right)$$

$$C = -\left[\frac{2}{R_1} + \frac{2}{R_2}\left(1 - \frac{2L}{R_1}\right)\right] \qquad D = -\left[\frac{2L}{R_1} - \left(1 - \frac{2L}{R_1}\right)\left(1 - \frac{2L}{R_2}\right)\right]$$

If travel n rounds

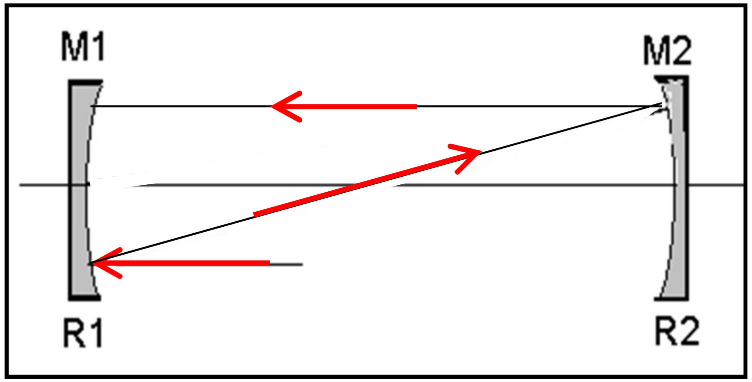
$$\begin{pmatrix} r_n \\ \theta_n \end{pmatrix} = \underbrace{T \cdots\cdots T}_{n} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = T_n \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

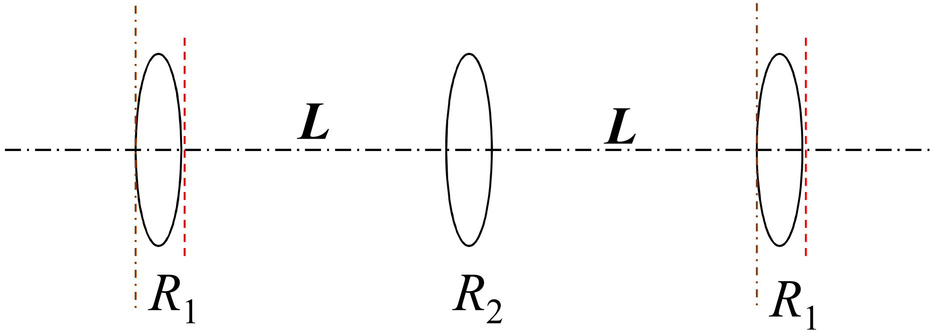
$$T_n = \frac{1}{\sin\varphi} \begin{pmatrix} A\sin n\varphi - \sin(n-1)\varphi & B\sin n\varphi \\ C\sin n\varphi & D\sin n\varphi - \sin(n-1)\varphi \end{pmatrix}$$

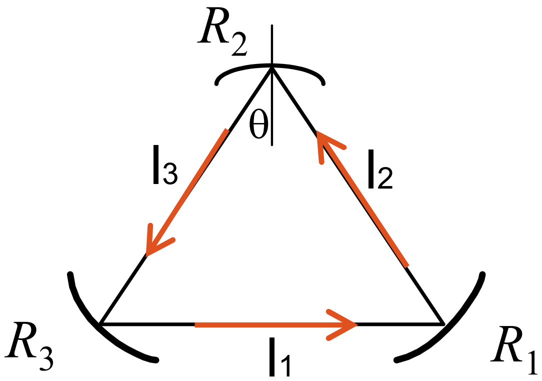
where $\varphi = \arccos \frac{1}{2}(A+D)$

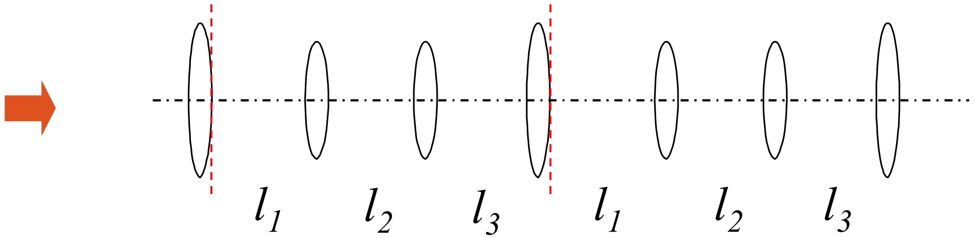
- ➢ Light travel in cavity can be treated as light pass through series of optical components
- ➢ The matrix is not related to their starting position and light travel direction
- ➢ **General Steps** (1) find the optical loop (2) fix the period (3) write down each component

Spherical cavity = series of lens



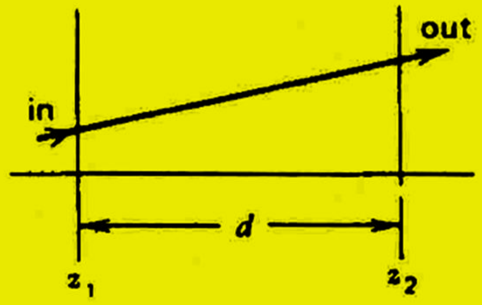
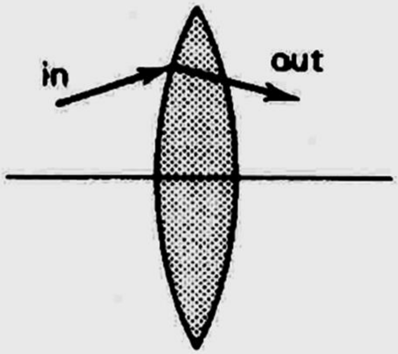
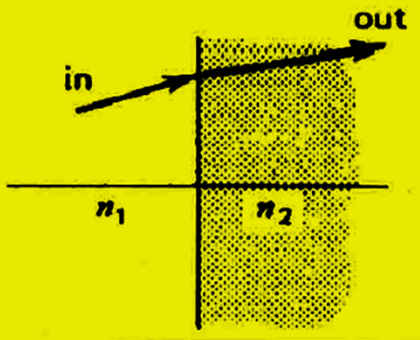






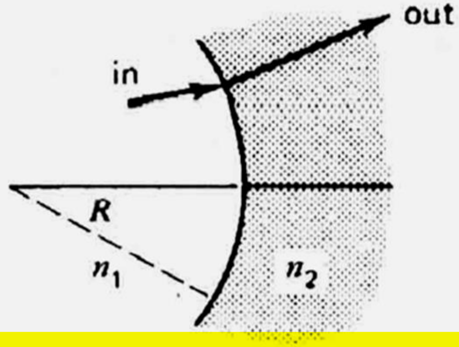
$$\begin{pmatrix} 1 & 0 \\ -2/R_3 & 1 \end{pmatrix}\begin{pmatrix} 1 & l_3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2/R_2 & 1 \end{pmatrix}\begin{pmatrix} 1 & l_2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2/R_1 & 1 \end{pmatrix}\begin{pmatrix} 1 & l_1 \\ 0 & 1 \end{pmatrix}$$

Ray matrices for some common cases

(1) Homogeneous Medium: Length d		$\begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix}$
(2) Thin Lens: Focal length f ($f>0$, converging; $f<0$, diverging)		$\begin{bmatrix} 1 & 0 \\ \frac{-1}{f} & 1 \end{bmatrix}$
(3) Dielectric Interface: Refractive indices n_1, n_2		$\begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix}$

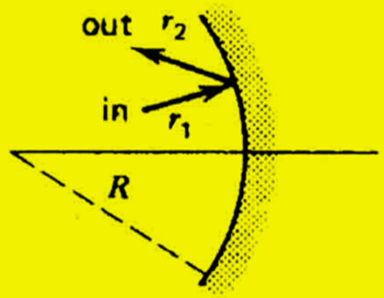
Ray matrices for some common cases

(4) Spherical Dielectric Interface:
Radius R



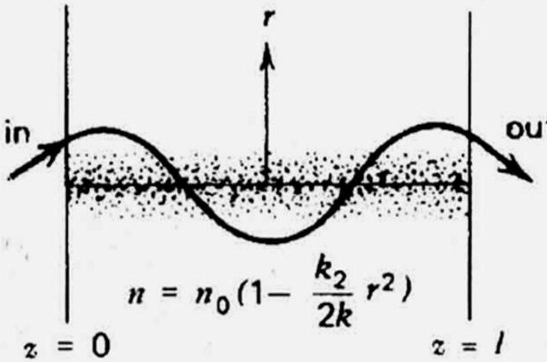
$$\begin{bmatrix} 1 & 0 \\ \frac{n_2-n_1}{n_2}\frac{1}{R} & \frac{n_1}{n_2} \end{bmatrix}$$

(5) Spherical Mirror:
Radius of curvature R



$$\begin{bmatrix} 1 & 0 \\ \frac{-2}{R} & 1 \end{bmatrix}$$

(6) A medium with a quadratic index profile

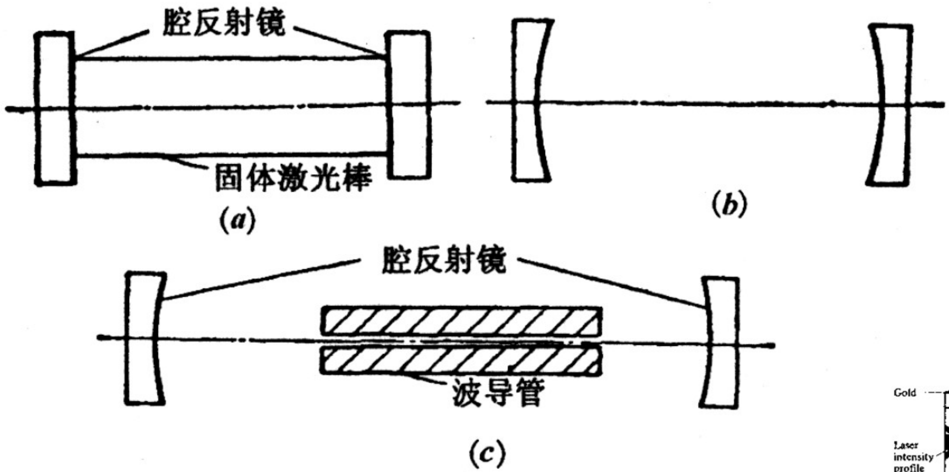


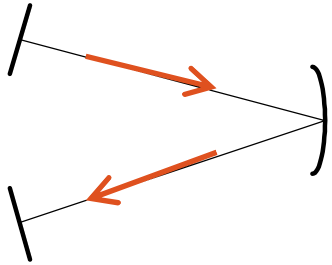
$$\begin{bmatrix} \cos\left(\sqrt{\frac{k_2}{k}}\,l\right) & \sqrt{\frac{k}{k_2}}\sin\left(\sqrt{\frac{k_2}{k}}\,l\right) \\ -\sqrt{\frac{k_2}{k}}\sin\left(\sqrt{\frac{k_2}{k}}\,l\right) & \cos\left(\sqrt{\frac{k_2}{k}}\,l\right) \end{bmatrix}$$

平方折射率介质

Classification of Optical Cavity

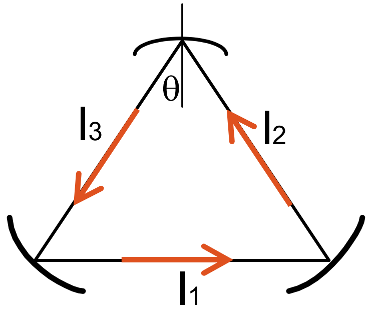
1. Construction

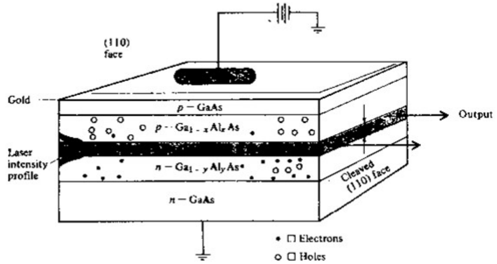




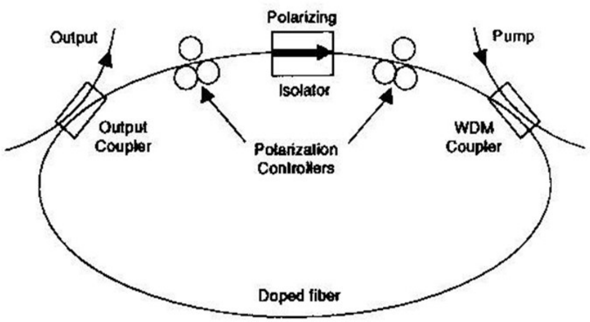
折叠腔

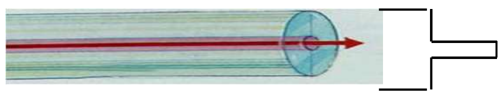
环形腔



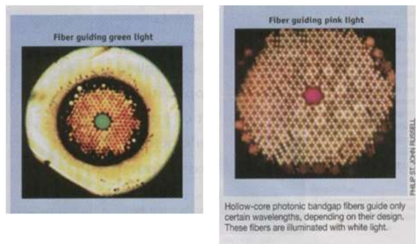


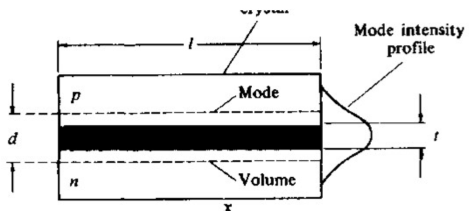
dielectric waveguide





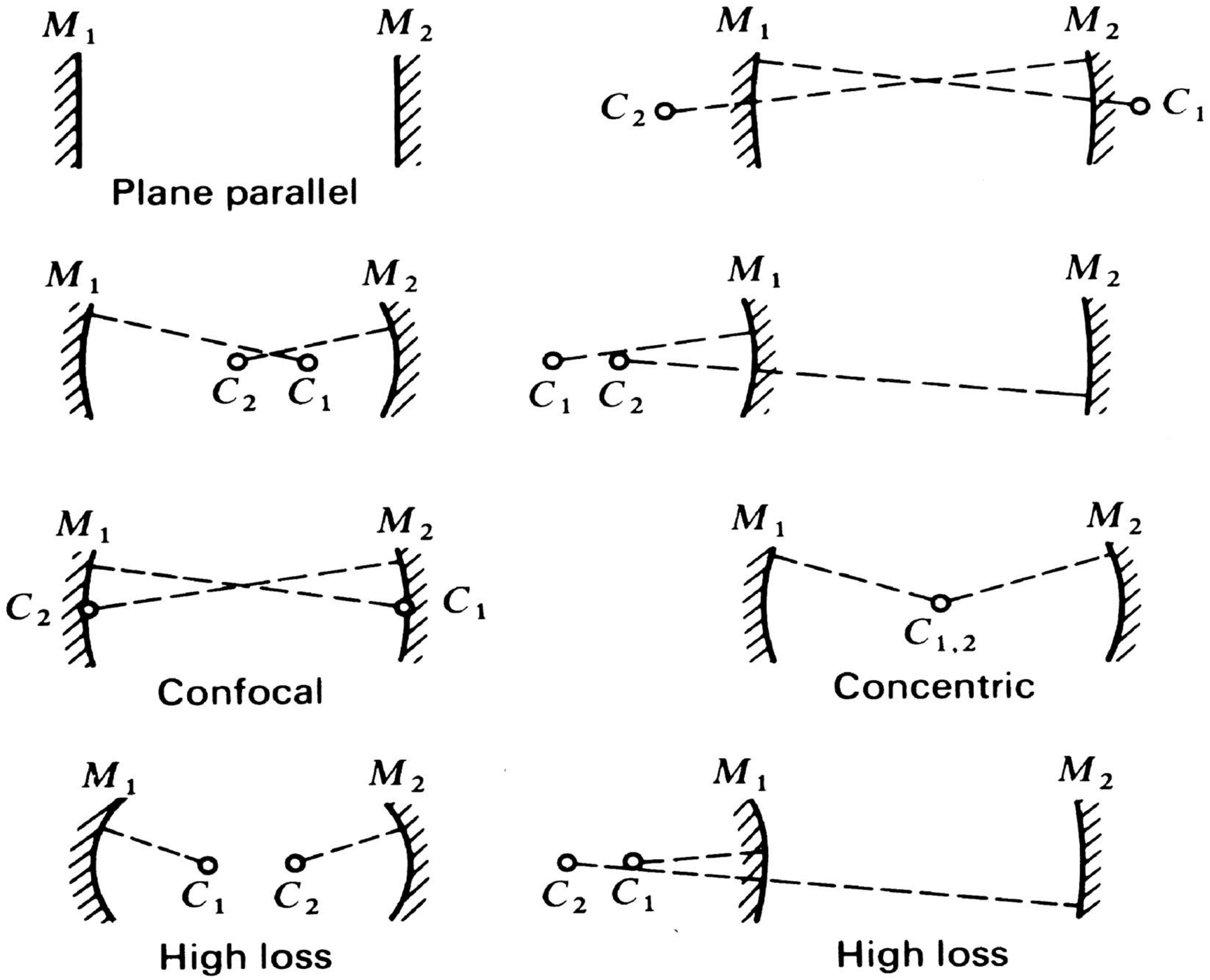
semiconductor laser





$\eta_2 > \eta_1$,
$\eta_2 > \eta_3$

2. Geometrical leakage loss: stable, semi-stable, critical



3. Stable condition for spherical mirror cavity

➢ Co-axis spherical mirror cavity: two mirrors are spherical mirror, have the same optical axis.

concave mirror $R > 0$; convex mirror $R < 0$; plane mirror $R = \infty$

凹透镜 凸透镜 平面镜

➢ Stable Condition: Geometrical loss

<u>Stable Cavity</u>: Any light with small angle can be reflected all the times inside the cavity and can not escape from it ⇔ small geometrical loss (low loss cavity)

<u>Unstable Cavity</u>: Light with small angle will escape out of the cavity after certain periods ⇔ large geometrical loss (high loss cavity)

➢ Ray matrix can be used to analyze the geometrical loss and determine whether the cavity is stable

If travel n rounds

$$\begin{pmatrix} r_n \\ \theta_n \end{pmatrix} = \underbrace{T \cdots\cdots T}_{n} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = T_n \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \begin{pmatrix} r_0 \\ \theta_0 \end{pmatrix}$$

$$T_n = \frac{1}{\sin\varphi} \begin{pmatrix} A\sin n\varphi - \sin(n-1)\varphi & B\sin n\varphi \\ C\sin n\varphi & D\sin n\varphi - \sin(n-1)\varphi \end{pmatrix}$$

where $\varphi = \arccos\frac{1}{2}(A+D)$

For stale cavity $\Rightarrow$ $\boxed{\frac{1}{2}|A+D| < 1}$

A_n B_n C_n D_n is infinite, then light can not escape from the cavity after reflected for n times. ϕ is a real number, and not equal to $k\pi$

Discussion $\varphi = \arccos\frac{1}{2}(A+D)$

(1) $\left|\frac{1}{2}(A+D)\right| < 1 \Rightarrow \varphi$ Real An, Bn, Cn, Dn is limited $\Rightarrow$ Stable

(2) $\left|\frac{1}{2}(A+D)\right| > 1 \Rightarrow \varphi$ Image $n\uparrow \rightarrow An, Bn, Cn, Dn\uparrow$ $\Rightarrow$ Unstable

(3) $\left|\frac{1}{2}(A+D)\right| = 1 \quad \Leftrightarrow \frac{1}{2}(A+D) = \pm 1 \Rightarrow \quad \varphi = k\pi \quad \begin{cases} k \text{ odd } (-1) \\ k \text{ even } (+1) \end{cases}$

Find the limitation

k odd $\quad \frac{1}{2}(A+D) = -1 \quad T_n = (-1)^{n+1}\begin{pmatrix} An+(n-1) & Bn \\ Cn & Dn+(n-1) \end{pmatrix}$

k even $\quad \frac{1}{2}(A+D) = +1 \quad T_n = 1^{(n+1)}\begin{pmatrix} An-(n-1) & Bn \\ Cn & Dn-(n-1) \end{pmatrix}$

Another Expression

$$\because\ A = 1 - \frac{2L}{R_2} \qquad D = -\left[\frac{2L}{R_1} - \left(1 - \frac{2L}{R_1}\right)\left(1 - \frac{2L}{R_2}\right)\right]$$

$$g_1 = 1 - L/R_1$$

$$g_2 = 1 - L/R_2$$

$$\frac{1}{2}(A + D) \equiv 1 - \frac{2L}{R_1} - \frac{2L}{R_2} + \frac{2L^2}{R_1 R_2}$$

$$-1 < \frac{1}{2}(A + D) < 1 \Rightarrow 0 < \left(1 - \frac{L}{R_1}\right)\left(1 - \frac{L}{R_2}\right) < 1 \Rightarrow \boxed{0 < g_1 g_2 < 1}$$

不是general表达

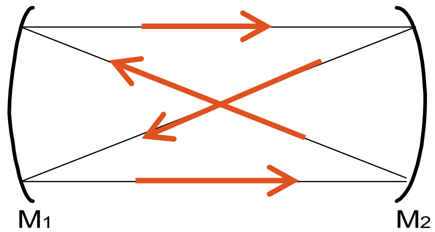
Stable Cavity	$0 < g_1 g_2 < 1$
Unstable Cavity	$g_1 g_2 > 1, \quad g_1 g_2 < 0$
Critical Cavity	$g_1 g_2 = 0, \quad g_1 g_2 = 1$

Some Examples

1. Planer mirror cavity ($R = \infty$)

$$T_n = \begin{pmatrix} 1 & 2Ln \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{cases} r_n = r_0 + 2Ln\theta_0 \\ \theta_n = \theta_0 \end{cases} \Rightarrow \begin{cases} \theta_0 \neq 0 \quad r_n \uparrow \quad \text{Unstable} \\ \theta_0 = 0 \quad r_n = r_0 \quad \text{Stable} \end{cases}$$

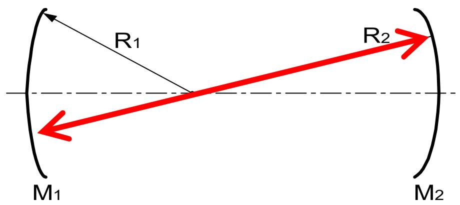
2. Symmetrical confocal cavity R_1=R_2=L



$$T_n = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Form close loop after two period: Stable

3. Concentric cavity R_1+R_2=L



Critical

$$\left|\frac{1}{2}(A+D)\right| < 1 \ \text{Stable;} \quad \left|\frac{1}{2}(A+D)\right| > 1 \ \text{Unstable;} \quad \left|\frac{1}{2}(A+D)\right| = 1 \ \text{Criticle}$$

It is used for all kinds of cavity. You can determine the stability by the ray matrix.